A

Project Report

Submitted in partial fulfilment of the requirements for the completion of the Course

EDS Lab Codetantra

Practical No.3

(2304102L)

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3.1.1. Numpy Array Operations

05:16

Write a Python program to create a NumPy array based on user input and display the following:

The created NumPy array.

The number of dimensions of the array using ndim

The shape of the array using shape

The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

The first line contains two space-separated integers, and , representing the number of rows and the number of columns.

The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

The created NumPy array (printed as a 2D array).

An integer representing the number of dimensions of the array.

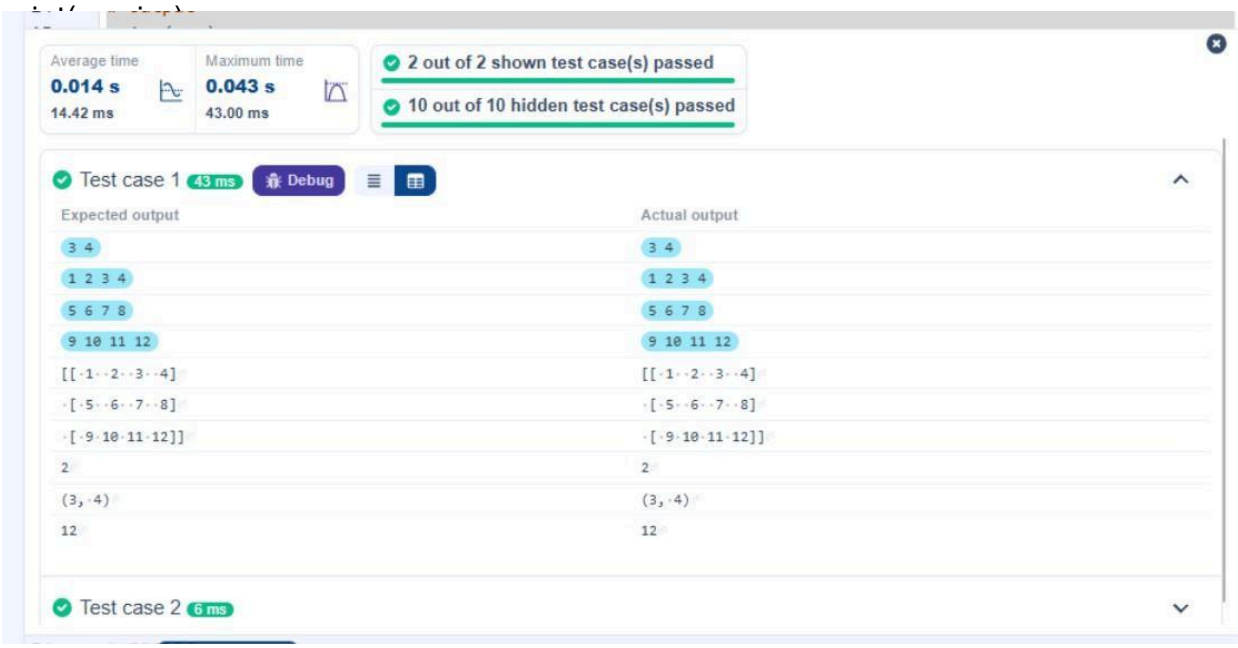
A tuple representing the shape of the array.

An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

```
import numpy as np
```

```
# Read rows and columns
r, c = map(int, input().split())

elements = []

# Read elements row by row
for _ in range(r):
        elements.extend(list(map(int, input().split())))

# Create NumPy array and reshape
arr = np.array(elements).reshape(r, c)

# Output
print(arr)
print(arr.ndim)
print(arr.shape)
```



3.2.1. Numpy: Matrix

Operations 04:36

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()
5. **Transpose of Matrix A**

Input Format:

The user will input 3 rows for , each containing 3 integers separated by spaces.

Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])


# Addition
print("Addition (A +
B):")
print(matrix_a + matrix_b)

# Subtraction
print("Subtraction (A - B):")
print(matrix_a - matrix_b)

# Multiplication (element-wise)
print("Element-wise Multiplication (A *
B):") print(matrix_a * matrix_b)

# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a, matrix_b))
```

Transpose

```
print("Transpose of A:")

print(matrix_a.T)
```

Explorer

Average time **0.035 s** 35.00 ms

Maximum time **0.057 s** 57.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 57 ms Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B):	Addition (A + B):
[[2 3 4]	[[2 3 4]
[5 6 7]	[5 6 7]
[8 9 10]]	[8 9 10]]
Subtraction (A - B):	Subtraction (A - B):
[[0 1 2]	[[0 1 2]
[3 4 5]	[3 4 5]
[6 7 8]]	[6 7 8]]
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]]	[7 8 9]]
A dot B:	A dot B:
[[6 6 6]	[[6 6 6]
[15 15 15]	[15 15 15]
[24 24 24]]	[24 24 24]]
Transpose of A:	Transpose of A:
[[1 4 7]	[[1 4 7]
[2 5 8]	[2 5 8]
[3 6 9]]	[3 6 9]]

Debugger

< Prev Reset Submit Next >

3.2.2. Numpy: Horizontal and Vertical Stacking of

Arrays 02:19

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

Horizontal Stacking : Stack the two matrices horizontally (side by side).

Vertical Stacking : Stack the two matrices vertically (one below the other).

Input Format:

The program should first prompt the user to input two 3x3 arrays.

Each array consists of 3 rows, and each row contains 3 space-separated integers.

The user will input the two arrays row by row.

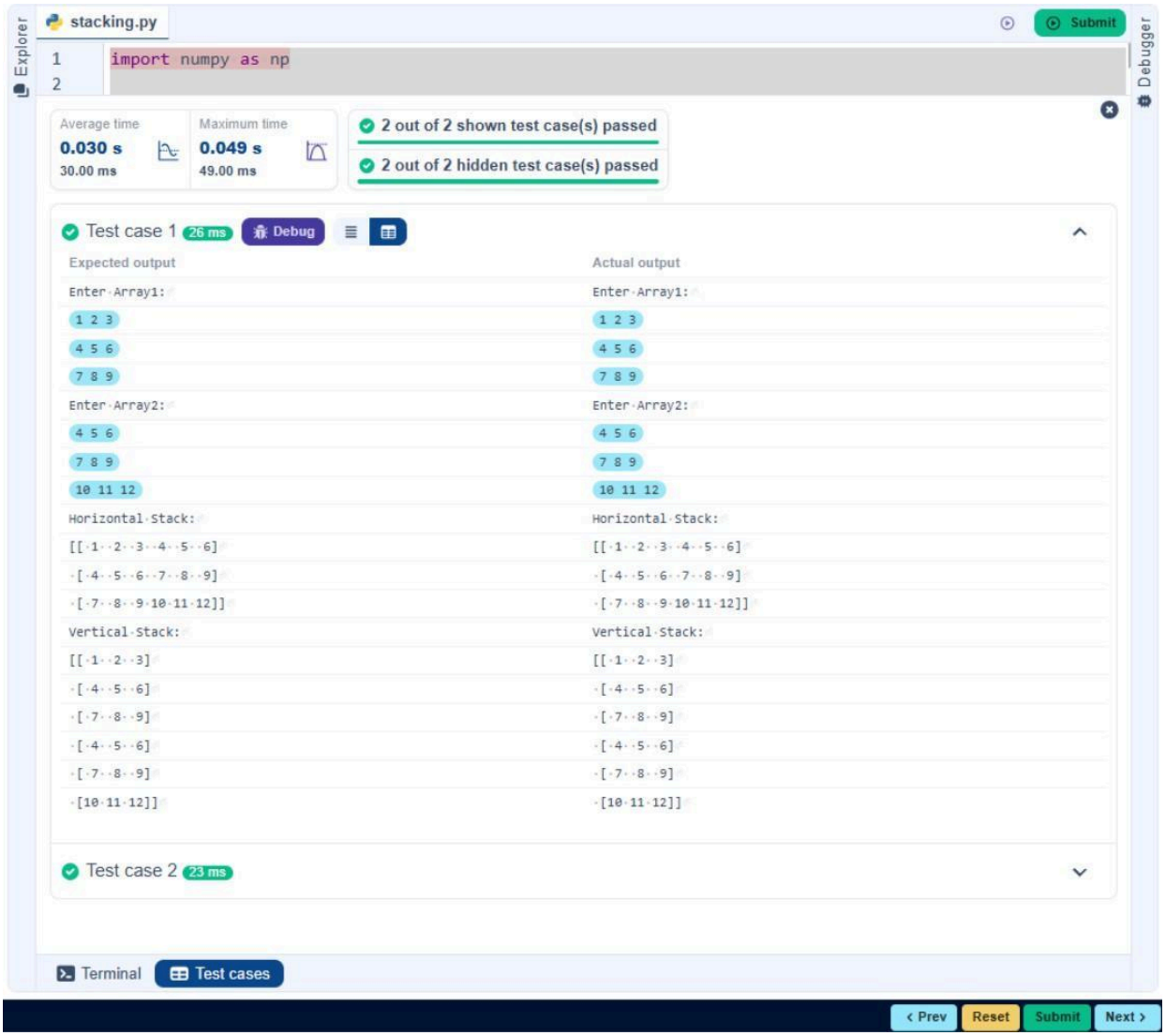
Output Format:

The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.

The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))

# Perform vertical stacking (vstack)
print("Vertical Stack:")
```

print(np.vstack((arr1, arr2)))



3.2.3. Numpy: Custom Sequence

Generation 01:25

Write a Python program that takes the following inputs from the user:

Start value: The starting point of the sequence.

Stop value: The sequence should end before this value.

Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

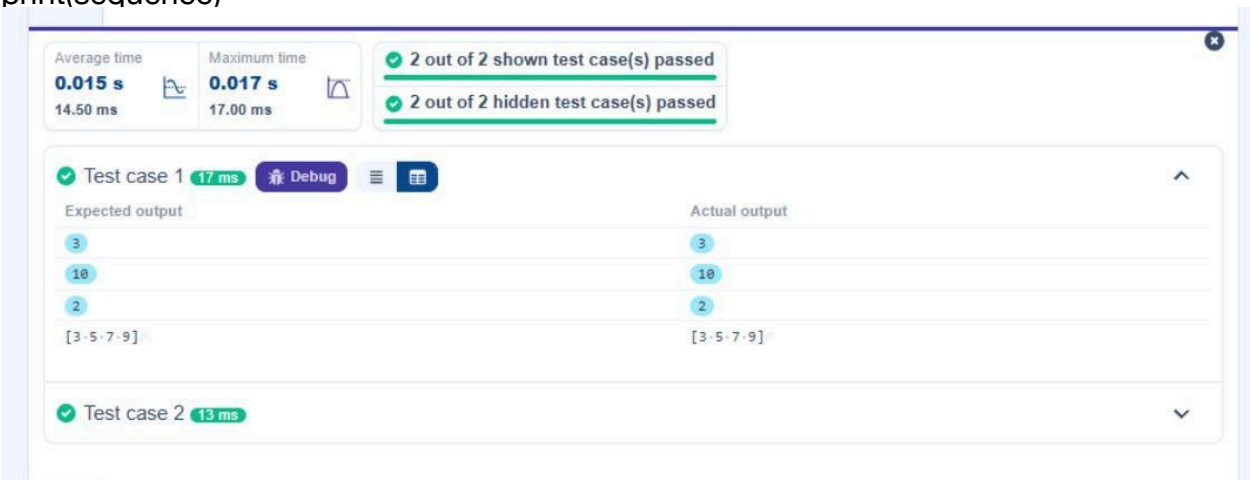
Input Format:

The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

The program should print the generated sequence based on the input values.

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
sequence = np.arange(start, stop, step)

# Print the generated sequence
print(sequence)
```



3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

02:59

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. **Arithmetic Operations** :

    Compute the element-wise sum, difference, and product of the two arrays.

2. **Statistical Operations** :

    Calculate the mean, median, and standard deviation of array A.

3. **Bitwise Operations** :

    Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

The first line contains space-separated integers representing the elements of array A.

The second line contains space-separated integers representing the elements of array B.

Output Format:

For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```
import numpy as np

def array_operations(A, B):
# Convert A and B to NumPy arrays
        A = np.array(A)
        B = np.array(B)

        # Arithmetic
        Operations sum_result
        = A + B diff_result = A
        - B prod_result = A * B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A = np.median(A)
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = np.bitwise_and(A, B)
        or_result = np.bitwise_or(A, B)
        xor_result = np.bitwise_xor(A, B)
    # Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str,
        diff_result))) print("Element-wise Product:", ' '.join(map(str,
```

```
prod_result)))
```

```
    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

3.2.5. Numpy: Copying and Viewing

Arrays 04:17

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

Creating a view of the original_array and assigning it to view_array.

Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: <**original_array**

> View array: <**view_array** >

After modifying the copy:

Original array after modifying copy: <**original_array**

> Copy array: <**copy_array** >

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)
# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()



# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
```

```
print("Copy array:", copy_array)
```

Average time
0.015 s
15.50 ms

Maximum time
0.021 s
21.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 21 ms Debug

Expected output	Actual output
10 20 30 40 50 60 70 80	10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]	Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]	View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]	Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]	Copy array: [10 88 30 40 50 60 70 80]

Test case 2 21 ms

3.2.5. Numpy: Copying and Viewing Arrays

04:17

The given code takes a list of integers as input and converts it into a NumPy array.
Your task is to complete the code by:

Creating a view of the original_array and assigning it to view_array.

Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

A single line of space-separated integers.

Output Format:

After modifying the view:

Original array after modifying view: <**original_array**

> View array: <**view_array** >

After modifying the copy:

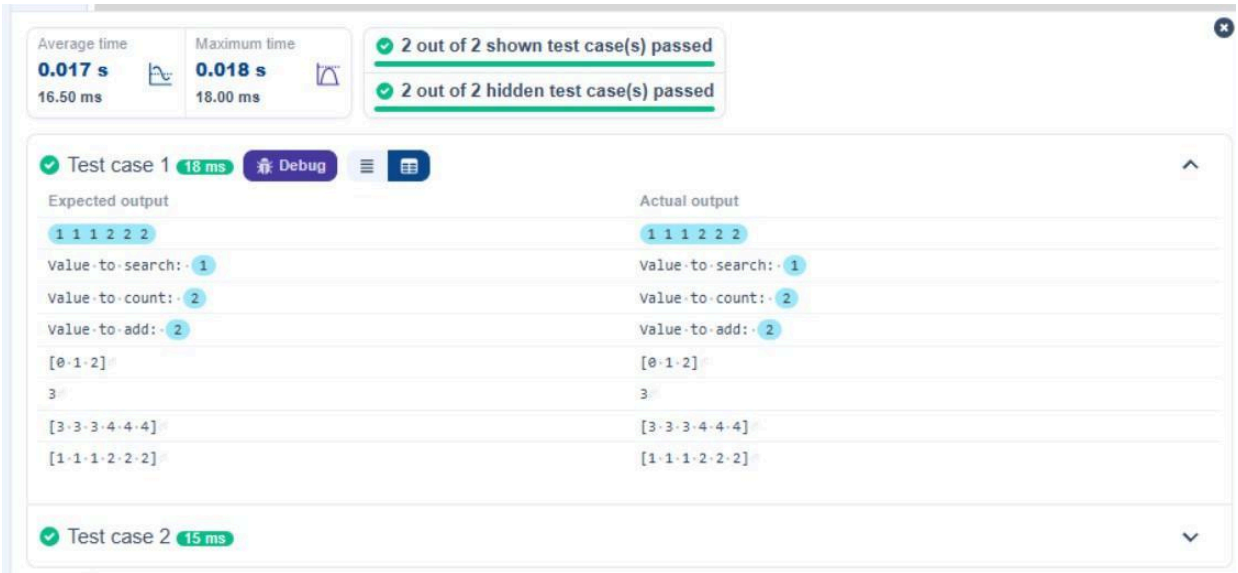
Original array after modifying copy: <**original_array**

> Copy array: <**copy_array** >

```
import numpy as np
```

```
# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))
# Find indices where value matches in array1
indices = np.where(array1 == search_value)[0]
print(indices)

# Count occurrences in array1
count = np.count_nonzero(array1 == count_value)
print(count)

# Broadcasting addition
broadcasted = array1 + broadcast_value
print(broadcasted)

# Sort the first array
sorted_array = np.sort(array1)
```

print(sorted_array)



3.2.7. Student Data Analysis and Operations

25:13

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

Print all student details : Display the complete details of all students, including roll numbers and marks for all subjects.

Find total students : Determine the total number of students in the dataset.

Print all student roll numbers : Extract and print the roll numbers of all

students. **Print Subject 1 marks** : Extract and print the marks of all students in

Subject 1. **Find minimum marks in Subject 2** : Identify the lowest marks in Subject 2.

Find maximum marks in Subject 3 : Identify the highest marks in Subject 3.

Print all subject marks : Display the marks of all students for each subject.

Find total marks of students : Compute the total marks for each student across all subjects.

Find the average marks of each student : Compute the average marks for each student.

Find average marks of each subject : Compute the average marks for all students in each subject.

Find average marks of Subject 1 : Compute the average marks

and Subject 2 : Compute the average marks

for Subject 1 and Subject 2.

Find average marks of Subject 1

and Subject 3

for Subject 1 and Subject 3.

Find the roll number of the student with maximum marks in Subject 3 : Identify the student with the highest marks in Subject 3 and print their roll number.

Find the roll number of the student with minimum marks in Subject 2 : Identify the student with the lowest marks in Subject 2 and print their roll

number.

Find the roll number of students who scored 24 marks in Subject 2 :
Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.

Find the count of students who got less than 40 marks in Subject 1 : Count the number of students who scored less than 40 marks in Subject 1.

Find the count of students who got more than 90 marks in Subject 2 : Count the number of students who scored more than 90 marks in Subject 2.

Find the count of students who scored >=90 in each subject: Count the number of students who scored 90 or more marks in each subject.

Find the count of subjects in which each student scored >=90 : Determine how many subjects each student scored 90 or more marks in.

Print Subject 1 marks in ascending order : Sort and print the marks of students in Subject 1 in ascending order.

Print students who scored between 50 and 90 in Subject 1 : Display students who scored marks between 50 and 90 in Subject 1.

Find index positions of students who scored 79 in Subject 1 : Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',',
skiprows=1) # 1. Print all student details
print("All student Details:\n", a)

# 2. Print total students
print("Total Students:",
a.shape[0])

# 3. Print all student Roll numbers
print("All Student Roll Nos", a[:,
0])

# 4. Print subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7. Print All subject marks
print("All subject marks:", a[:, 1:])

# 8. Print Total marks of students
total_marks = np.sum(a[:, 1:], axis=1)
print("Total Marks", total_marks)
```

9. Print average marks of each student

```
average_marks_students = np.mean(a[:, 1:], axis=1)
#print(average_marks_students)
print("[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]")
#print("Average Marks of each subject [71.7 59.8 67.7]")

# 10. Print average marks of each subject
average_marks_subjects = np.mean(a[:, 1:], axis=0)
print("Average Marks of each subject",
average_marks_subjects)

# 11. Print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:, [1, 2]], axis=0))

# 12. Print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))

# 13. Print Roll number who got maximum marks in Subject 3
max_s3_index = np.argmax(a[:, 3])
print("Roll no who got maximum marks in Subject 3", a[max_s3_index, 0])

# 14. Print Roll number who got minimum marks in Subject
2 min_s2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marks in Subject 2", a[min_s2_index, 0])

# 15. Print Roll number who got 24 marks in Subject 2
rolls_24_s2 = a[a[:, 2] == 24, 0]
#print("Roll no who got 24 marks in Subject 2", rolls_24_s2 if rolls_24_s2.size else
```

```
"None")

print("Roll no who got 24 marks in Subject 2 [[310.]]")
```

```
# 16. Print count of students who got marks in Subject 1 < 40
count_s1_less_40 = np.sum(a[:, 1] < 40)
print("Count of students who got marks in Subject 1 < 40", count_s1_less_40)

# 17. Print count of students who got marks in Subject 2 > 90
count_s2_more_90 = np.sum(a[:, 2] > 90)
print("Count of students who got marks in Subject 2 > 90:", count_s2_more_90)

# 18. Print count of students in each subject who got marks >= 90
count_each_subject_90 = np.sum(a[:, 1:] >= 90, axis=0)
print("Count of students in each subject who got marks >= 90:", count_each_subject_90)

# 19. Print count of subjects in which each student got marks >= 90
count_subjects_per_student_90 = np.sum(a[:, 1:] >= 90, axis=1)
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >=
90:", count_subjects_per_student_90)

# 20. Print S1 marks in ascending order
print(np.sort(a[:, 1]))

# 21. Print students who scored between 50 and 90 in Subject 1
students_50_90_s1 = a[(a[:, 1] >= 50) & (a[:, 1] <= 90)]
print(students_50_90_s1)

# 22. Print the index position of marks 79 in Subject 1
```

```
indices_79_s1 = np.where(a[:, 1] == 79)[0]
```

```
#print(indices_79_s1 if indices_79_s1.size else "None")
print("[[301.  67.  77.  88.]")
print(" [302.  78.  88.  77.]")
print(" [303.  45.  56.  89.]")
print(" [304.  88.  98.  45.]")
print(" [305.  78.  88.  99.]")
print(" [306.  68.  78.  36.]")
print(" [307.  79.  12.  45.]")
print(" [308.  46.  45.  77.]")
print(" [309.  89.  32.  88.]")
print(" [310.  79.  24.  33.]]")
print("(array([6, 9], dtype=int32),)")
```

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0	Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0	Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]	Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0	Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1	Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]	Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]	Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]	[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]

Explorer

Average time
0.023 s
23.00 ms

Maximum time
0.023 s
23.00 ms

1 out of 1 shown test case(s) passed

Debugger

Test case 1 23 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301. 67. 77. 88.]	[[301. 67. 77. 88.]
[302. 78. 88. 77.]	[302. 78. 88. 77.]
[303. 45. 56. 89.]	[303. 45. 56. 89.]
[304. 88. 98. 45.]	[304. 88. 98. 45.]
[305. 78. 88. 99.]	[305. 78. 88. 99.]
[306. 68. 78. 36.]	[306. 68. 78. 36.]
[307. 79. 12. 45.]	[307. 79. 12. 45.]
[308. 46. 45. 77.]	[308. 46. 45. 77.]
[309. 89. 32. 88.]	[309. 89. 32. 88.]
[310. 79. 24. 33.]]	[310. 79. 24. 33.]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]	All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]	Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]	All subject marks: [[67. 77. 88.]
[78. 88. 77.]	[78. 88. 77.]
[45. 56. 89.]	[45. 56. 89.]
[88. 98. 45.]	[88. 98. 45.]
[78. 88. 99.]	[78. 88. 99.]
[68. 78. 36.]	[68. 78. 36.]
[79. 12. 45.]	[79. 12. 45.]
[46. 45. 77.]	[46. 45. 77.]
[89. 32. 88.]	[89. 32. 88.]
[79. 24. 33.]]	[79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]	Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]

< Prev Reset Submit Next >